

Xinyang Wu

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Profile

- **Software Engineer** with an MSc from **ETH Zürich**, shipping production-grade **Python / FastAPI** systems with **full-stack delivery** (React / Next.js), **observability** (OpenTelemetry, Datadog) and **CI/CD / release-gate discipline**.
- Hands-on **ML / GenAI engineering**: production **agentic LLM services**, **RAG-style** grounding, evaluation harnesses, **experiment tracking** (**Weights & Biases**), dataset versioning; **PyTorch**, **TensorFlow**, **scikit-learn**.
- From algorithmic / signal-processing roots through large-scale data pipelines on **multi-GPU Linux clusters** — comfortable translating research and stakeholder needs into reliable shipped systems.

Experience

Software Engineer, Miltenyi Biotec – Zürich, Switzerland

Sept 2024 – present

- Designed and optimized a **high-throughput data ingestion and processing platform** for biosensor time-series using **Python / FastAPI** implemented advanced indexing on the storage layer, reducing query latency by **10%** and supporting concurrent serving for **30** internal users.
- Architected and deployed a **production-grade agentic LLM service** featuring **asynchronous streaming** and robust **session management**, improving automated spec-generation throughput by **35%**.
- Designed and implemented **CI/CD pipelines (GitLab CI, Docker)** to automate linting, testing, and **multi-stage containerized deployments** for both the FastAPI backend and Next.js frontend, cutting deployment cycle time by **10%** and saving the team **3 hours/week** of manual environment setup.
- Integrated **automated model evaluation and data validation** into the CI/CD infrastructure, enabling continuous integration of fine-tuned weights and guardrail verification before production rolling updates.
- Instrumented **FastAPI prediction endpoints** with **OpenTelemetry** metrics and structured logs, exporting telemetry via an **OTEL Collector** to **Datadog** built **RED dashboards** and **triage runbooks** for **p95/p99 latency** and **failure-rate** monitoring — cutting prediction-regression triage from **2 min to 30 s**.
- Deployed self-hosted internal **AI chat (OpenWebUI)** on Linux and built **evaluation + model-deployment tooling** around it; React / TypeScript front-end kept lightweight as an auxiliary delivery layer.

Machine Learning Engineer Intern, BOSCH – Stuttgart, Germany

Mar 2024 – Aug 2024

- Engineered **scalable dataset versioning** and **automated data-pipeline tooling** on a **distributed multi-GPU Linux cluster**, decreasing model-training data-prep time by **50%** and ensuring **100% reproducibility** for downstream computer-vision stacks.
- Developed a **distributed compute pipeline (Python, PyTorch)** to prototype **3D video reconstruction with Gaussian Splatting**, optimizing **GPU memory utilization by 30%** during large-scale spatial-temporal upsampling.
- Automated data-pipeline provisioning and container validation on the cluster via **CI/CD automation (Docker)**, ensuring fully reproducible training environments for cross-team computer-vision workloads.
- Work productized into **first-author patent US 19/402,148** (EU/CN counterparts).

Research Assistant (Generative AI), ETH Zürich — Computer Vision Lab – Zürich, Switzerland

Mar 2023 – Sept 2023

- **Fine-tuned Stable Diffusion** (Hugging Face Diffusers, PyTorch) on a shared multi-GPU cluster; built **distributed training and evaluation pipelines** for foundation-model experimentation.

Data Analyst Intern, Seventh Affiliated Hospital, Sun Yat-sen University – Shenzhen, China

Aug 2022 – Sept 2022

- Biomedical signal processing (EMG / EEG): wavelet denoising and feature extraction in Python / MATLAB.

Selected Publications, Patents & Projects

- **First-Author Patent (US 19/402,148)**: *Gaussian Flow for Video Perception and Temporal Dataset Upsampling* — distributed spatial-temporal upsampling algorithm for large-scale video data. Counterparts filed in EU and CN.
- **First-Author Paper (under review, PHIRO, top journal for medical physics)**: *Real-Time 4D-MRI Reconstruction with Implicit Neural Representations* — cut reconstruction MAE ~46% and diaphragm motion error ~49% vs. prior baselines; experiments orchestrated and tracked end-to-end on **Weights & Biases**.
- **Co-Inventor, Patent CN110363755A**: *Cardiac MRI/CT Segmentation and Detection*. (First-author paper, BDCPS 2021.)
- **Project — AI Reading Partner** (Google Chrome Built-in AI Challenge, 2024): shipped a **multimodal agentic** Chrome extension powered by the **Gemini API** — DOM extraction → backend → live UI, a compact **RAG-shaped** reference application demonstrating end-to-end GenAI integration.

Education

ETH Zürich, MSc in Information Technology and Electrical Engineering – Zürich, Switzerland

Sept 2022 – June 2025

- Coursework: **Probabilistic AI**, **Big Data for Engineers**, Image Analysis & Computer Vision, Dynamic Programming and Optimal Control.

Sun Yat-sen University, BEng in Biomedical Engineering – Shenzhen, China

Sept 2018 – June 2022

Skills

Languages & Front-End: Python, SQL, TypeScript / JavaScript, Java, C, Bash, MATLAB; React, Next.js, Material-UI for full-stack delivery.

Backend & Distributed Systems: FastAPI, REST APIs, asynchronous streaming, session management, indexing & query optimization, full-text retrieval, high-throughput data pipelines; multi-GPU Linux clusters, distributed training & inference, GPU memory optimization, NumPy.

ML & GenAI: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers / Diffusers, foundation-model fine-tuning, evaluation harnesses (MAE / RMSE / Dice, regression), agentic LLM & RAG applications, LLM APIs (Gemini, OpenAI-compatible).

Observability, CI/CD & MLOps: OpenTelemetry, OTEL Collector, Datadog, RED dashboards, p95/p99 latency & failure-rate monitoring, runbooks; Docker, GitHub Actions, GitLab CI, container-based deployment, model deployment & serving; Weights & Biases experiment tracking, dataset versioning, reproducible environments, validation gates, performance profiling.

Languages

- English (C1), Japanese N1 (C1), Mandarin Chinese (Native), German (B1).